A GIS BASED SOIL MOISTURE MAPPING FOR THOOTHUKUDI DISTRICT

A PROJECT REPORT

Submitted by

MAHENDRAPALLAVAN G (953221135024)

NANTHAKUMAR P (953221135027)

OLIVIZHI K (953221135028)

in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

in

GEOINFORMATICS ENGINEERING

UNIVERSITY VOC COLLEGE OF ENGINEERING, THOOTHUKUDI

ANNA UNIVERSITY: CHENNAI 600 025

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BONAFIDE CERTIFICATE

Certified that this project report **" A GIS BASED SOIL MOISTURE MAPPING FOR THOOTHUKUDI DISTRICT"** is the bonafide work of "**MAHENDRAPALLAVAN G (953221135024), NANTHAKUMAR P (953221135027), OLIVIZHI K (953221135028)"** who carried out the project work under my supervision.

Dr. S. RAJAKUMAR, M.E, Ph.d	Dr. K. NELSIA PRIYA DHARSINI, M.E, Ph.d
HEAD OF THE DEPARTMENT	**SUPERVISOR**
Assistant Professor	Assistant Professor
Department of Civil Engineering	Department of Civil Engineering
University VOC College of Engineering	University VOC College of Engineering
Thoothukudi – 628008	Thoothukudi - 628008

Submitted for the VIVA-VOICE Examination held on……

INTERNAL EXAMINER EXTERNAL EXAMINER

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ABSTRACT

Soil moisture is a critical factor influencing agricultural productivity, hydrological modeling, and climate studies. Traditional methods of soil moisture measurement are often limited by spatial and temporal constraints. To address these limitations, this study leverages Google Earth Engine (GEE), a powerful cloud-based platform for processing geospatial data, to map soil moisture variations in the Tuticorin district. By utilizing remotely sensed data, including satellite imagery and climate datasets, we propose an innovative approach for the estimation and monitoring of soil moisture across large areas. This research integrates multiple data sources, including the European Space Agency's Soil Moisture and Ocean Salinity (SMOS) and NASA's Soil Moisture Active Passive (SMAP) satellite products, alongside rainfall data, vegetation indices, and land surface temperature. The study aims to provide a more accurate, real-time assessment of soil moisture levels, contributing to effective water resource management and enhancing agricultural practices in the region. The results demonstrate the potential of Google Earth Engine as a tool for large-scale environmental monitoring and decision-making.

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LIST OF ABBREVIATION

ARCGIS	Arc Geographic Information System
CSV	Comma-Separated Values
GEE	Google Earth Engine
GEOTIFF	Georeferenced Tagged Image File Format
GIS	Geographical Information System
GLDAS	Global Land Data Assimilation System
ISSCS	Indian Standard Soil Classification System
MODIS	Moderate Resolution Imaging Spectroradiometer
NASA	National Aeronautics and Space Administration
NDVI	Normalized Difference Vegetation Index
QGIS	Quantum Geographic Information System
RMSE	Root Mean Square Error
RS	Remote Sensing
RVI	Radar Vegetation Index
SAR	Synthetic Aperture Radar
SMI	Soil Moisture Index
SMAP	Soil Moisture Active Passive
SMOS	Soil Moisture and Ocean Salinity
SRTM	Shuttle Radar Topography Mission
SSM	Surface Soil Moisture
SPI	Normalized Difference Vegetation Index
VH	Vertical to horizontal
VV	Vertical to vertical

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Soil is assumed as an inexhaustible resource that is used continually for increasing agricultural production. However, the truth is that soil has turned lifeless in places with increasing development and industrialization. Soil has the potential to drastically impact a country's economy and environment .Therefore, Soil moisture measurement is very important. Soil moisture is the amount water held in soil pores within the top layer(5-10cm)from the Earth's surface. Monitoring soil moisture is essential for precision agriculture, irrigation management, drought prediction, and climate studies. Understanding soil moisture helps optimize water usage, improve crop yield, and mitigate environmental issues like soil erosion and land degradation.

The soil moisture plays a very important role in controlling and regulating the interaction between the land surface and atmosphere. The quantification of soil moisture can open a whole new spectrum of interpretation and analysis of data which in turn, lead to greater efficiency and deeper insights. The findings of this study will contribute to better water resource management, optimized irrigation planning, and improved agricultural productivity in the Tuticorin regions.

1.1.1 APPLICATION OF GIS AND REMOTE SENSING

A Geographical Information System (GIS) is a computer-based tool for mapping and analyzing features events on the Earth. GIS technology integrates common database operation, such as query and statistical analysis with maps. The GIS is used to study

geologist features, analyze soils and strata, assess seismic information, and or create three-dimensional display of geographical features.

Remote sensing is the art and science of making measurements of the earth using sensors on airplanes or satellites. These sensors collect data in the form of images and provide specialized capabilities for manipulating. analyzing, and visualizing those images. Remote sensed imagery is integrated within a GIS. GIS manages location-based information and provides tools for display and analysis of various statistics, including population characteristics, economic development opportunities, vegetation types.

1.1.2 GIS AND REMOTE SENSING IN SOIL MAPPING

Soil mapping is an essential aspect of sustainable land management, agriculture, and environmental monitoring. Conventional soil surveys are usually costly and time-consuming, involving intensive fieldwork. With the advancements in Geographic Information Systems (GIS) and Remote Sensing (RS), soil mapping has been transformed to deliver cost-saving, efficient, and high-resolution spatial information.

Among remote sensing technology, the European Space Agency's (ESA) radar-based Earth observation satellite Sentinel-1 has a very important role in soil mapping. Sentinel-1, unlike optical sensors, functions in the C-band Synthetic Aperture Radar (SAR) frequency and is able to capture data regardless of weather or cloud cover. It is, therefore, well-suited for estimating soil moisture, analyzing texture, and land surface classification. By combining Sentinel-1 SAR data with GIS, scientists and policymakers can examine soil characteristics across large regions, track changes over time, and create predictive models of soil well-being and fertility. The use of GIS spatial analysis procedures combined with remote sensing-based soil data facilitates precision agriculture, erosion management, and sustainable land use planning.

1.2 SOIL

Soil is an accumulation of loose material from mechanical and chemical weathering of rocks and containing a large mixture of various organic substances on the Earth's surface. The mineral material from which a soil forms is called parent material. Rock, whether its origin is igneous, sedimentary, or metamorphic, is source of all soil mineral materials. Soil contains mineral particles, organic matter, water and air. The combination of these determines the soil's properties its texture, structure, porosity, chemistry and color. Soil is made up of different sized particles. In the Indian Standard Soil Classification System (ISSCS), soils are classified into groups according to size and are further divided into coarse, medium and fine sub groups. The grain size range is used as the basis for grouping soil particles into boulder, cobble, gravel, sand, silt or clay. Gravel, sand, silt and clay are represented by group symbols G, S, M, and C respectively. Physical weathering produces very coarse and coarse soils. Chemical weathering produces generally fine soils.

1.2.1 SOIL MOISTURE

Soil Moisture is a key variable in controlling the exchange of water and heat energy between the land surface and the atmosphere through evaporation and plant transpiration. Soil moisture content ismeasured in $m^{3.}$ The volumetric soil moisture content and vegetation cover.

1.2.2 WATER HOLDING CAPACITY

Water holding capacity is the total amount of water a soil can hold at field capacity.

The clay soil had the highest water holding capacity because clay particles are so tiny and have many small pore spaces that make water move slower and the sandy soil

have good drainage but low water and nutrient holding capacities.

1.2.3 ROLE OF SOIL

- Soils serve as media for growth of all kinds of plants. Soils modify the atmosphere by emitting and absorbing gases like carbon dioxide, methane, water vapor and dust.

- Soils provide habitat for animals and micro-organism that live in the soil, that account for most of the living things on Earth.

- Soils absorb, hold, alter, and purify most of the water in terrestrial systems.

- Soils process recycled nutrients, including carbon, so that living things can use them over and over again.

- Soils serve as engineering media for construction of foundations, roadbeds, dams and buildings, and preserve or destroy artifacts of human endeavor.

1.3 OBJECTIVES

- To map soil moisture content using google earth engine for Thoothukudi District (Oct to Dec 2024).

- To identify low and high moisture zone for agriculture monitoring.

1.4 NEED FOR THE STUDY

Soil moisture plays an important role in the development of weather patterns and precipitation. Soil moisture map helps to identify surface water content and is used for

early warning of droughts, irrigation scheduling, and crop yield forecasting for the benefits of people in a region.

CHAPTER 2

LITERATURE REVIEW

2.1 GENERAL

The given literatures are important thing for gathering the knowledge about this project titled as "A GIS-BASED SOIL MOISTURE MAPPING FOR TUTICORIN DISTRICT". Then the following contents are all about the important terms under the project. The relevant literature related to mapping of soil moisture has been complied and presented in the following section.

2.2 STUDY ON SOIL MOISTURE MAPPING USING GOOGLE EARTH ENGINE

Aliihsan Şekertekin, Aycan Murat Marangoz , Saygın Abdikan, "SOIL MOISTURE MAPPING USING SENTINEL-1A SYNTHETIC APERTURE RADAR DATA" Vol: 5 Issue:2 July 2018. This study focuses on estimating and mapping soil moisture distribution using C-band Synthetic Aperture Radar (SAR) data from the Sentinel-1A satellite, acquired on 24 April 2016. The research was conducted in an agricultural region of Bergama, İzmir, where in-situ soil moisture measurements were collected from 20 test fields during the satellite pass. The relationship between soil moisture, local incidence angle, and the SAR backscattering coefficient was analyzed. A multiple regression model was developed to estimate soil moisture, achieving high accuracy with an R^2 of 0.84 and RMSE of 2.46%. The results demonstrated that Sentinel-1A SAR data is effective for soil moisture mapping.

NazmusSazib, Iliana Mladenova and John Bolten "LEVERAGING THE GOOGLE EARTH ENGINE FOR DROUGHT ASSESSMENT USING GLOBAL SOIL

MOISTURE DATA" 11 August 2018, This study focuses on evaluating two global soil moisture datasets and web-based tools developed for agricultural drought monitoring and crop forecasting using Google Earth Engine (GEE). The datasets combine satellite observations from SMOS and SMAP missions with a modified Palmer model using a data assimilation approach. The tools allow users to analyze soil moisture variability, drought duration, and intensity, and compare them with other drought indicators. Assessments in South Africa and Ethiopia showed strong agreement between soil moisture anomalies, NDVI, and SPI. The integration of these datasets and tools into GEE enhances accessibility, especially for users in developing countries, supporting better drought risk assessment and early warning systems.

Sachin Sutariya, Ankur Hirapara, Momin Meherbanali, M.K. Tiwari, Vijay Singh, Manik H. Kalubarme." SOIL MOISTURE ESTIMATION USING SENTINEL-1 SAR DATA AND LAND SURFACE TEMPERATURE IN PANCHMAHAL DISTRICT, GUJARAT STATE", Vol: 8 Issue:1 25 Nov 2020. This study explores the potential of retrieving soil moisture using Sentinel-1 C-band SAR data and Land Surface Temperature (LST) derived from LANDSAT-8 data. Sentinel-1 and LANDSAT-8 data from February 2020 were used, alongside soil samples collected from 70 test fields across three talukas. SAR data were processed to estimate backscattering coefficients, while LST was calculated using thermal bands from LANDSAT-8. Soil Moisture Index (SMI) was computed using LST values. Regression analysis showed that VV polarization was more sensitive to soil moisture than VH polarization. Multiple regression analysis yielded good accuracy with R^2 values around 0.78, allowing reliable prediction of soil moisture across the study area.

Felix Greifeneder , Claudia Notarnicola and Wolfgang Wagner," A MACHINE LEARNING-BASED APPROACH FOR SURFACE SOIL MOISTURE

ESTIMATIONS WITH GOOGLE EARTH ENGINE ” 27 May 2021. This study presents a machine learning-based approach for high-resolution (50 m) soil moisture mapping by integrating Landsat-8 optical and thermal data, Sentinel-1 SAR images, and modeled data, executed on Google Earth Engine. Unlike existing medium-resolution (1 km) products, this method offers finer detail, particularly beneficial for agriculture. The model is trained using globally distributed in-situ data from the International Soil Moisture Network. Validation results showed high accuracy with an RMSE of 0.04 m^3/m^3 and R^2 of 0.81. Additionally, the study introduces a Python package and cloud-based framework that simplifies data collection and processing without requiring local downloads.

Jinyang Du , John S. Kimball , Rajat Bindlish , Jeffrey P. Walker , Jennifer D. Watts,” LOCAL SCALE (3-M) SOIL MOISTURE MAPPING USING SMAP AND 2 PLANET SUPER DOVE ” June 10,2022. This study proposes a new method to estimate high-resolution (3-m) surface soil moisture (SSM) by combining multi-sensor data fusion, machine learning, and CDF matching techniques. It integrates Planet SuperDove imagery with NASA's SMAP data to bridge the gap between coarse and fine spatial scales. The method demonstrated improved accuracy, reducing bias and RMSE by ~0.01 cm^3/cm^3 over the Yanco region in Australia, while maintaining SMAP’s frequent temporal sampling and sensitivity. The approach effectively captured local variations in soil moisture linked to vegetation and irrigation patterns, offering potential benefits for water management, precision agriculture, and disaster response.

Janani N, Balaji Kannan, Nagarajan K, Thiyagarajan G, Duraisamy M R,” TREND ANALYSIS AND VARIABILITY OF SATELLITE-BASED SOIL MOISTURE DATA FOR THE LOWER BHAVANI BASIN, TAMIL NADU USING GOOGLE EARTH ENGINE” May 11,2023. This study analyzes soil moisture trends in the Lower Bhavani basin, Tamil Nadu, from 2003 to 2022 using satellite-based GLDAS data accessed via

Google Earth Engine. The analysis showed higher soil moisture during monsoon months and lower levels in summer. Trend analysis using the Man-Kendall test and Sen's slope revealed an increasing soil moisture trend during winter (October to February) from 2003-2012, and during both winter and monsoon seasons from 2013-2022. No decreasing trend was observed. These findings can aid in better monitoring and management of the environmental and hydrological systems in the basin.

Abolghasem Akbari, MajidRajabi Jagharg Azizan AbuSamah Jonathan PeterCox|, Mojtaba Gholamzadeh , Alireza Araghi, PatriciaM.Saco, "UTILIZATION OF THE GOOGLE EARTH ENGINE FOR THE EVALUATION OF DAILY SOI TEMPERATURE DERVED FROM GLOBAL LAND DATA ASSIMILATION SYSTEM IN TWO DIFFERENT DEPTHS OVERASE MIARID REGION" 27 June 2024. This study evaluates the performance of GLDAS-Noah soil temperature (ST) data against observed data from 13 synoptic stations in a semi-arid region of Iran using Google Earth Engine (GEE). ST data were analyzed over 5 years at two soil depths (0–10 cm and 40–100 cm) for daily minimum, maximum, and average temperatures. The results showed strong correlations, with high efficiency metrics across both layers, though a negative bias was observed, leading to underestimation of ST. Despite this bias, GLDAS-Noah data proved useful for soil temperature estimation in arid regions with limited observations. Additionally, GEE demonstrated its effectiveness for large-scale geospatial analysis.

Shadia A. Alzurqani Shawn Jackson , Hamdi A. Zurqani , Don White Jr., Kathleen Bridges," GOOGLE EARTH ENGINE APPLICATION FOR MAPPING AND MONITORING DROUGHT PATTERNS AND TRENDS: A CASE STUDY IN ARKANSAS, USA", 19 Oct 2024. This study focuses on analyzing drought patterns in Arkansas, USA, using remotely sensed indices to assess their impact on vegetation and develop a cloud-based framework for drought monitoring. It investigates the correlation

between Vegetation Health Index (VHI) and meteorological drought indices and applies Mann-Kendall trend analysis to identify drought trends. Results show that drought is most common during March and August, with about 31% of the area experiencing varying drought intensities, especially affecting agricultural regions like the Mississippi Alluvial Valley Plain. The study offers valuable insights for policymakers and stakeholders to manage and mitigate drought impacts effectively.

Zhenghao Li, Qiangqiang Yuan & Xin Su," HIGH-SPATIAL-RESOLUTION SURFACE SOIL MOISTURE RETRIEVAL USING THE DEEP FOREST MODEL IN THE CLOUD ENVIRONMENT OVER THE TIBETAN PLATEAU",15 Feb 2024. This study aims to retrieve high-resolution (10 m) surface soil moisture over the Tibetan Plateau using Sentinel-1 SAR, Sentinel-2 multispectral data, and auxiliary datasets like land cover and soil texture. A cloud-based workflow integrating Google Earth Engine, Colab, and Drive was developed, utilizing a machine learning model called Deep Forest. The modified Deep Forest model showed superior performance with an R^2 of 0.834 and low error metrics. It outperformed other models in both cross-validation and site-based validation. The cloud-based framework effectively streamlined the entire process, offering a practical approach for environmental management and similar studies.

CHAPTER 3

MATERIALS AND METHODS

3.1 GENERAL

The present study attempts to map soil moisture content for South Tamil Nadu. The details of data analyses and techniques adopted are described in the following section. On reviewing the literature, the methodology for the study area is drafted. The methodology involves data collection, radiometric correction ,soil moisture estimation and finally mapping on soil moisture content thoothukudi district.

3.2 DESCRIPTION OF STUDY AREA

Thoothukudi , A southern district of Tamil Nadu and also known as "Pearl City" due to the pearl fishing carried out in the town. Thoothukudi lies on the Coromandel coast of the Bay of Bengal. It covers area of 353.07 km^2 (136.32 sq mi) and had a population of 500,000 in 2024. Thoothukudi is located at 8.7642ºN 78.1348ºE. Thoothukudi is located in South India, on the Gulf of Mannar, about 604 kilometers (375 miles) south of Chennai.

3.2.2 CLIMATE

Thoothukudi experiences a tropical savanna climate characterized by sweltering summers, hot winters, and occasional heavy rain during the northeast monsoon.

3.2.3 SOIL

Thoothukudi district has coastal sand found along the shoreline. Red soil sand is present in inland areas. Alluvial and river sand occur in riverbeds, while some regions have dune sand due to wind deposition.

3.2.4 TEMPERATURE

Thoothukudi registers a maximum temperature of 39 °C (102 °F) in the month of May and a minimum temperature of 32 °C (90 °F) in the month of December. The city receives adequate rainfall only during the months of October and November.

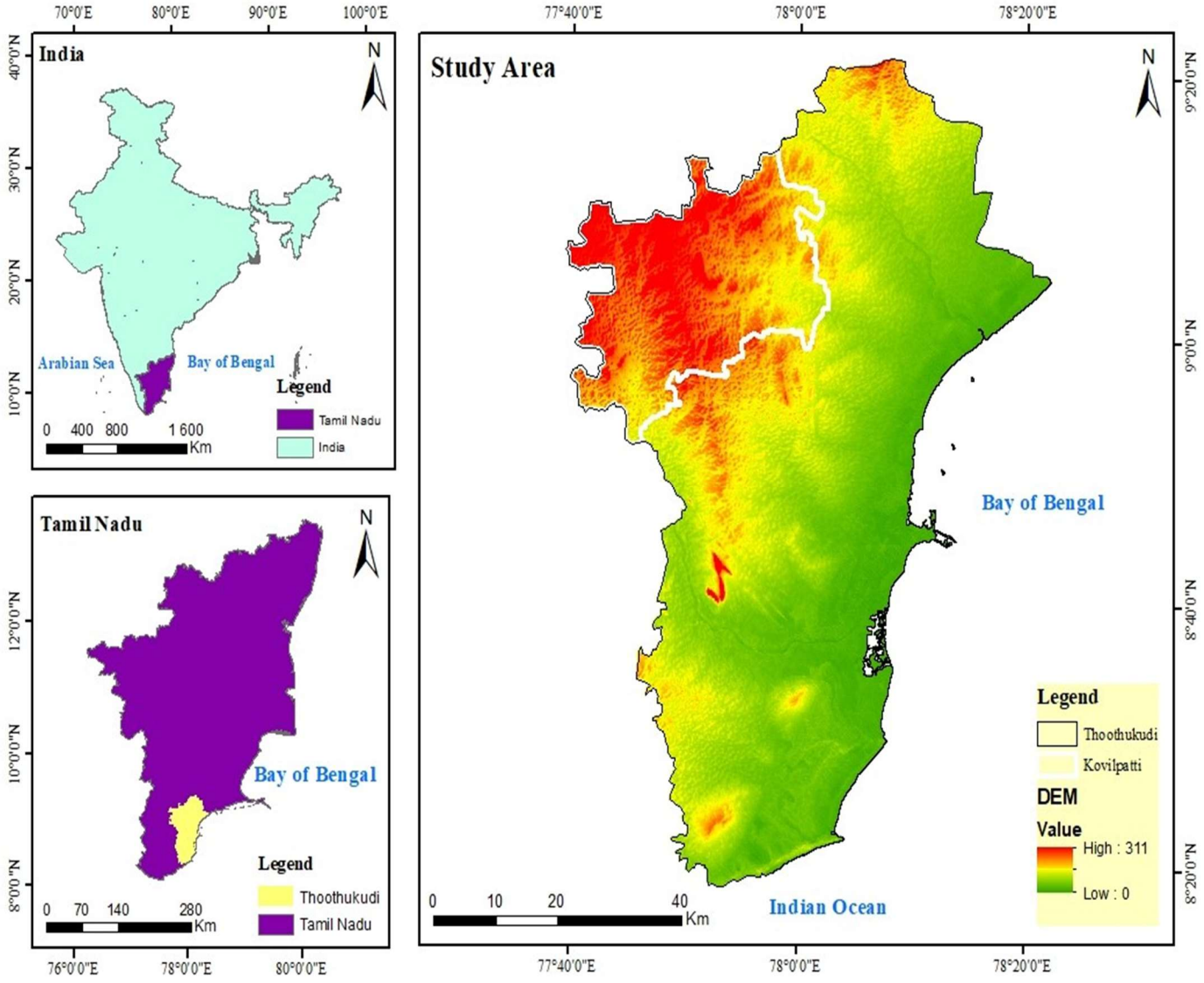
3.2.5 WATER BODIES

There are no large reservoirs in Thoothukudi district so, the Papanasam and Manimuthar dams located in the Tirunelveli district in the Thamirabarani River's flow are the main sources of irrigation. Other than the Thamirabarani river, the river Vaipar in Vilathikulam taluk, the river Karumeni which traverses through Satankulam, and Thiruchendur taluks, Palayakayal are all sources

3.3 LOCATION AND AREA

A Study area is Thoothukudi district. It is situated in the south-eastern corner of Tamil Nadu. Thoothukudi lies on 8.7642ºN latitude 78.1348ºE longitude. Thoothukudi is located in South India, on the Gulf of Mannar, about 604 kilometers (375 miles) south of Chennai.

Fig 3.1 Study area map



3.4 METHODOLOGY

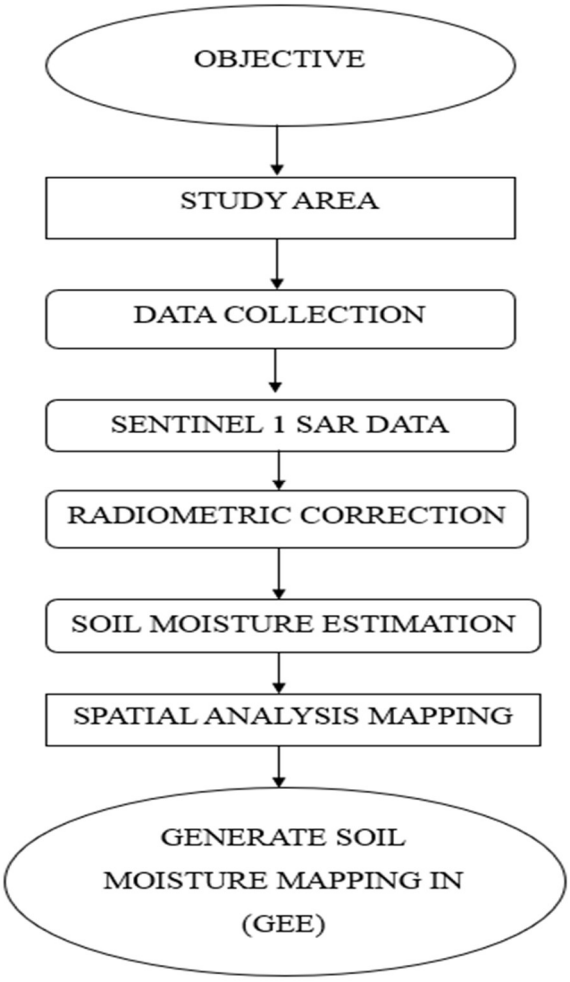


Fig 3.2 : Methodology

3.5 SOFTWARE USED

3.5.1 GOOGLE EARTH ENGINE

Google Earth Engine (GEE) is a cloud platform for geospatial computing and analysis, perfectly suitable for environmental tracking, such as soil moisture mapping.

3.5.2 APPLICATIONS IN THIS STUDY

Utilized Sentinel-1C SAR data, SAR backscatter processing, NDVI computation, and Soil Moisture Index (SMI) creation.

Created time-series charts and moisture maps.

3.5.3 PROGRAMMING LANGUAGE

JavaScript API utilized within the GEE Code Editor to script and visualize.

Python API employed optionally for automation and machine learning operations.

3.5.4 QGIS (QUANTUM GIS)

QGIS is an open-source free desktop GIS application employed for the editing, visualization, and analysis of spatial data.

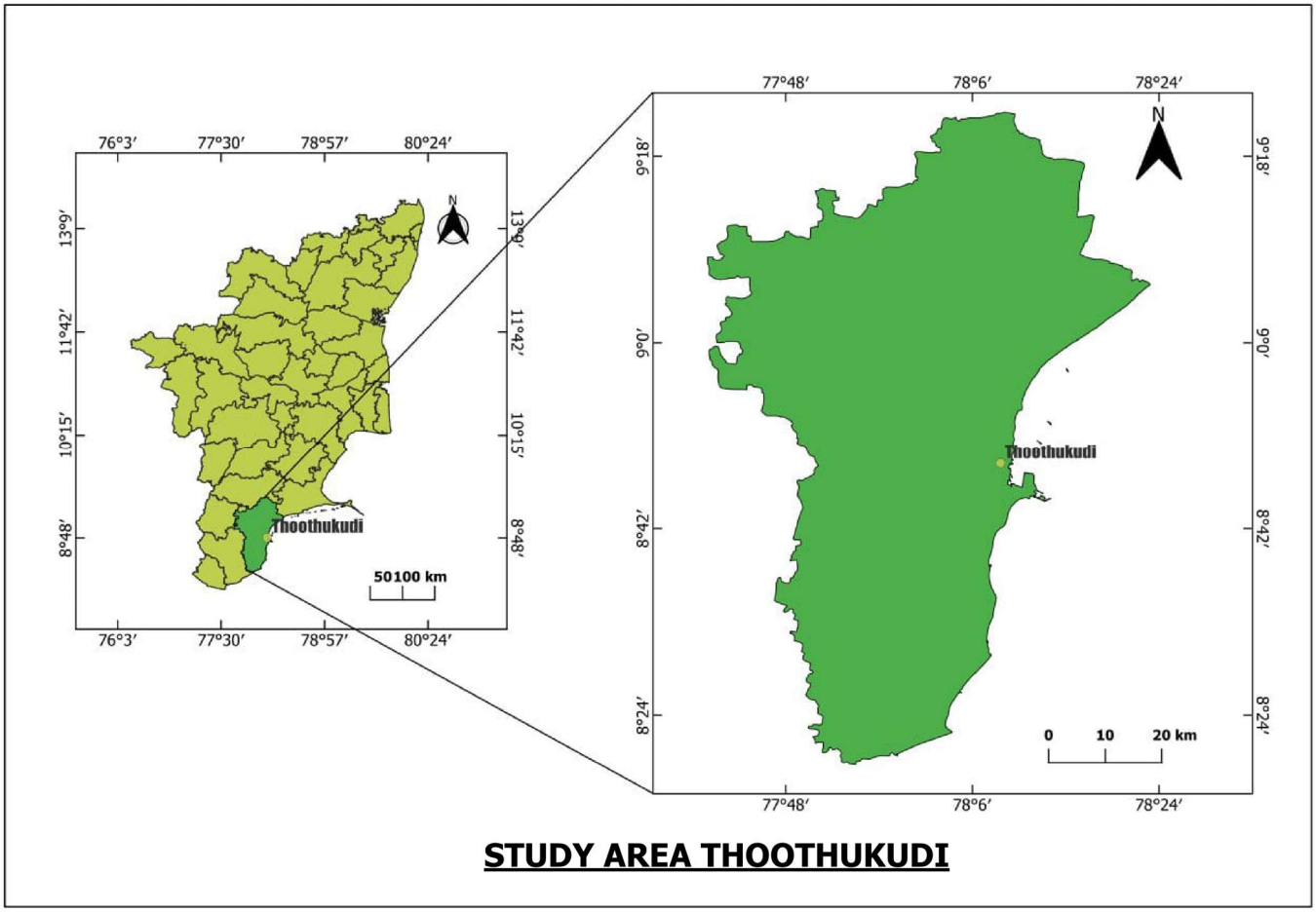


Fig 3.3 Study Area Thoothukudi

3.5.4.1 QGIS OVERVIEW

QGIS is an open-source and free Geographic Information System (GIS) software that allows users to create, visualize, analyze, and publish geospatial data in a broad variety of formats and platforms. It has been developed by an international community and sponsored by the Open Source Geospatial Foundation (OSGeo). QGIS has become one of the most powerful and versatile GIS tools available today. It is both raster and vector data capable, has real-time data editing capabilities, and gives users access to thousands of plugins that expand its functionality — including remote sensing tools, terrain analysis tools, spatial statistics tools, web mapping tools, and geocoding tools. QGIS is extremely compatible with numerous standard GIS data formats, including Shapefile, GeoTIFF, KML, CSV, and WMS, WFS, and PostGIS services. It is easily integrated with other open-source software such as GRASS GIS, SAGA GIS, and GDAL, making its processing capabilities even stronger. Installed on Windows, macOS, Linux, and Android (QField), QGIS finds applications in academia, government, environmental management, urban planning, agriculture, disaster risk assessment, and many more. Its simplicity in use, active development team, and lack of licensing fees make it the best option for both beginners and experts in the geospatial industry.

3.5.5 USES IN THIS STUDY

Importing and analyzing raster and vector layers. Reprojecting data and fixing geolocation. Executing spatial analysis such as zonal statistics, clipping, and classification. Employed for map preparation, symbology, and exporting high-quality maps for reporting purpose.

3.6 MOST IMPORTANT FEATURES FOR SOIL MOISTURE MAPPING

Semi-Automatic Classification Plugin (SCP) for moisture zones classification.

Raster Calculator for derivation of NDVI or soil moisture index from imported bands. Field calculator and attribute table for integration of moisturedata.

3.6.1 ARC GIS

ArcGIS is a commercial geographic information system (GIS) software developed by Esri, with advanced spatial analytics, 3D modeling, and data management capabilities.

3.6.1.0 ARCGIS OVERVIEW

ArcGIS is a full-featured and robust Geographic Information System (GIS) software suite created by Esri (Environmental Systems Research Institute). It is utilized globally by government, business, research, and environmental professionals to analyze, visualize, and manage spatial and geographic information. ArcGIS enables users to create interactive maps, conduct advanced spatial analysis, handle large geodatabases, and model spatial relationships. The platform comprises numerous tools and products like ArcGIS Pro (desktop software), ArcMap, ArcGIS Online (cloud GIS), and ArcGIS Enterprise (server-based solutions). Being based on remote sensing, geostatistics, 3D mapping, and real-time data integration, ArcGIS can tackle difficult real-world problems in domains like urban planning, natural resource management, public safety, and transportation. Its compatibility with formats such as shapefiles, GeoTIFF, and web services (WMS, WFS) renders it a versatile and necessary tool for spatial decision-making and digital cartography. Although it is a proprietary software that needs to be licensed, ArcGIS is greatly appreciated due to its powerful features, regular updates, and compatibility with emerging technologies like machine learning, big data, and IoT.

3.6.2 APPLICATION

Executed spatial interpolation (e.g., Kriging, IDW) to predict soil moisture at unsampled points.

Executed multi-criteria analysis for high/low moisture zones identification. Handled large geospatial data and produced professional grade cartographic outputs. Employed Arc Tool box for geoprocessing tasks like raster reclass, zonal statistics, and terrain analysis.

3.6.3 ADVANCED FEATURES EMPLOYED

Model Builder: Designed automated workflows for repeated soil moisture analysis processes. ArcGIS Online: Facilitated sharing and viewing interactive web maps.

3.6.4 ROLE OF SOFTWARES IN THIS PROJECT COMPARED

Software Main Use Strengths Limitation GEE Data access, preprocessing, analysis Cloud-based, scalable, free satellite access Requires coding, QGIS Visualization, map layout, spatial analysis Free, user-friendly, plugin-rich Processing is slow with large datasets.

3.7 PROGRAMMING LANGUAGE USED

A blend of programming languages has been utilized in this project to extract, analyze, and visualize soil moisture information. These languages support various geospatial analysis, data automation, and interfacing with external tools. The major languages utilized are JavaScript and Python, each playing different roles.

3.7.1 JAVASCRIPT (GOOGLE EARTH ENGINE CODE EDITOR)

JavaScript is the major language utilized in the Google Earth Engine (GEE) Code Editor, an environment based on the web browser.

3.7.2 APPLICATIONS IN THIS PROJECT

Writing scripts to load and filter Sentinel-1 SAR and Landsat imagery. Applying radiometric and terrain correction to SAR data. Creating vegetation indices like NDVI. Calculating the Soil Moisture Index (SMI).

Visualizing spatial and temporal data using maps and charts. Exporting processed results in Geo-TIFF, CSV, and SHP format.

3.7.3 PYTHON (GOOGLE EARTH ENGINE API, QGIS, ARCGIS)

Python is a general-purpose language with extensive use in automation, machine learning, and platform integration. It was used in combination with: Google Earth Engine Python API for complex processing. QGIS Python console (PyQGIS) for batch processing and plugin creation. ArcGIS Python scripting (ArcPy) for geoprocessing and modeling.

3.7.4 APPLICATIONS IN THIS PROJECT

Automating routine tasks such as downloading datasets or batch analysis. Developing custom soil moisture models with machine learning libraries like scikit-learn. Integrating GEE with external tools such as Google Colab or Jupyter Notebooks for interactive analysis and visualization.

Upscaling ArcGIS processes using Model Builder scripts and ArcPy geoprocessing.

3.8 NEED OF LANGUAGES

JavaScript Google Earth Engine Scripting, mapping, real-time visualization Python GEE, QGIS, ArcGIS Automation, data processing, machine learning Both languages provide powerful features for remote sensing analysis, data fusion, and custom tool development, and they are extremely useful in soil moisture mapping work.

3.8.1 GOOGLE EARTH ENGINE

Google Earth Engine (GEE) is a cloud computing platform for geospatial analysis and processing, best suited for environmental monitoring, like soil moisture mapping.

3.8.2 APPLICATIONS IN THIS STUDY

Retrieved Sentinel-1C SAR and Landsat data. Utilized SAR backscatter processing, NDVI computation, and Soil Moisture Index (SMI)calculation.
Compiled time-series charts and moisture maps.

3.8.3 PROGRAMMING LANGUAGE

JavaScript API utilized in the GEE Code Editor for scripting and visualization. Python API utilized optionally for machine learning and automation tasks

CHAPTER 4

STUDY AREA AND METHODS

4.1 DESCRIPTION OF STUDY AREA

Thoothukudi is a coastal district in the south-eastern region of Tamil Nadu, India. It is an important contributor to the state's economic, industrial, and agricultural activity. The district is famous for its port trade, salt industry, and dryland farming.

For the current study, Thoothukudi has been chosen for soil moisture mapping because it possesses varied soils, climatic conditions, and agricultural reliance on monsoonal rains and irrigation sources.

4.1.1 GEOGRAPHICAL LOCATION

Latitude: 8.7642° N

Longitude: 78.1348° E

Total Area: About 4,745 sq. km

Elevation: Relatively flat coastal plains with negligible undulating landscape.

Coastline: About 121 km along the Bay of Bengal (Gulf of Mannar).

Thoothukudi is bordered by Tirunelveli district to the west, Ramanathapuram to the north, and the Gulf of Mannar to the east. The administrative headquarters is the city of Thoothukudi.

4.2 CLIMATE AND RAINFALL

CLIMATE

Climate Type: Tropical savanna

Summer Temperature: Max. 39°C (May).

Winter Temperature: Approximately 26–32°C.

RAINFALL

Average Annual Rainfall: ~640–700 mm.

Received mostly during Northeast Monsoon (Oct–Dec).

Cyclonic storms occasionally affect rainfall patterns.

4.3 SOIL CHARACTERISTICS

Thoothukudi district contains a variety of soils that influence water holding and soil moistness very strongly:

Red soils : Inland areas; moderately permeable, applicable for dryland cultivation.

Coastal sandy soils : Along the seashore; low water holding capacity.

Alluvial soils : Along the river bed; very good for farming because fertile.

Black soils (small patches): High clay and water holding capacity but more susceptible to cracking under dry conditions.

These diverse soil types render the district an ideal candidate for spatial soil moisture analysis.

4.4 LAND USE AND VEGETATION

Agriculture: Predominant land use, with crops such as paddy, groundnut, cotton, black gram, and sorghum.

Forest cover: Limited (~5%), primarily degraded scrub and thorny vegetation.

Urbanization: Clustering around Thoothukudi city and along highways and coastal regions.

4.4.1 WATER RESOURCES

Thoothukudi does not have large perennial rivers but relies on:

Tamirabarani River: Through canal systems from Papanasam and Manimuthar

Dams (in Tirunelveli district).

Vaippar and Karumeniyar Rivers: Seasonal rivers facilitating irrigation in Vilathikulam and Sathankulam taluks.

Tanks and ponds: Traditional systems of local water storage and supply.

4.5 METHODOLOGY

The adopted methodology for soil moisture mapping of Thoothukudi district incorporates remote sensing methods, Google Earth Engine (GEE) processing, and GIS-based spatial analysis. It is developed to estimate surface soil moisture from satellite data and map spatial variability over the study region.

4.6 OVERVIEW

The main steps included are:

1) Satellite Data Collection.

2) Preprocessing of Remote Sensing Data.

3) Soil Moisture Estimation Using Remote Sensing Indices.

4) Validation and Accuracy Assessment (if ground truth data available).

5) Mapping and Analysis in GIS Environment.

4.7 SATELLITE DATA COLLECTION

1) Sentinel-1 (C-band Synthetic Aperture Radar)

2) Delivers backscatter intensity (VV and VH polarization).

3) Best suited for soil moisture estimation due to penetration through vegetation and cloud insensitivity.

4)Employed for computing NDVI (vegetation cover) and Land Surface Temperature (LST).

5) Other data sources: SMAP, MODIS, GLDAS (for comparison and cross-

validation if necessary).

6) All datasets were retrieved and processed via Google Earth Engine platform.

4.7.1 PREPROCESSING OF SATELLITE DATA

With the help of GEE, the following are the preprocessing tasks undertaken:

SAR DATA (SENTINEL-1):

1) Radiometric calibration

2) Terrain correction using SRTM DEM

3) Speckle filtering (optional)

4) Choice of VV polarization for soil moisture sensitivit

5) Cloud masking (using QA bands or FMask)

6) Conversion to surface reflectance

7) Derivation of indices (e.g., NDVI)

4.8 SOIL MOISTURE ESTIMATION METHODS

Various methods were employed for the estimation of soil moisture from remote sensing data:

1) Backscatter Analysis from Sentinel-1:

2) Radar backscatter coefficient ($\sigma°$) is related to soil moisture content.

3) Higher backscatter values generally represent higher surface wetness.

4) Regression models (multiple/linear) are employed to make moisture estimates.

4.8.1 MAPPING AND ANALYSIS IN GIS

After creating soil moisture layers in GEE:

1) Exported as GeoTIFF and imported into QGIS/ArcGIS.

2) Carry out reclassification into zones (e.g., High, Medium, Low moisture).

3) Run zonal statistics by land use or administrative boundary

CHAPTER 5

RESULT AND DISCUSSION

5.1 GENERAL

The present study attempts to map soil moisture content for South Tamil Nadu. The details of data analyses and techniques adopted are described in the following section. On reviewing the literature, the methodology for the study area is drafted. The methodology involves data collection, radiometric correction ,soil moisture estimation and finally mapping on soil moisture content thoothukudi district.

5.2 DATA COLLECTION

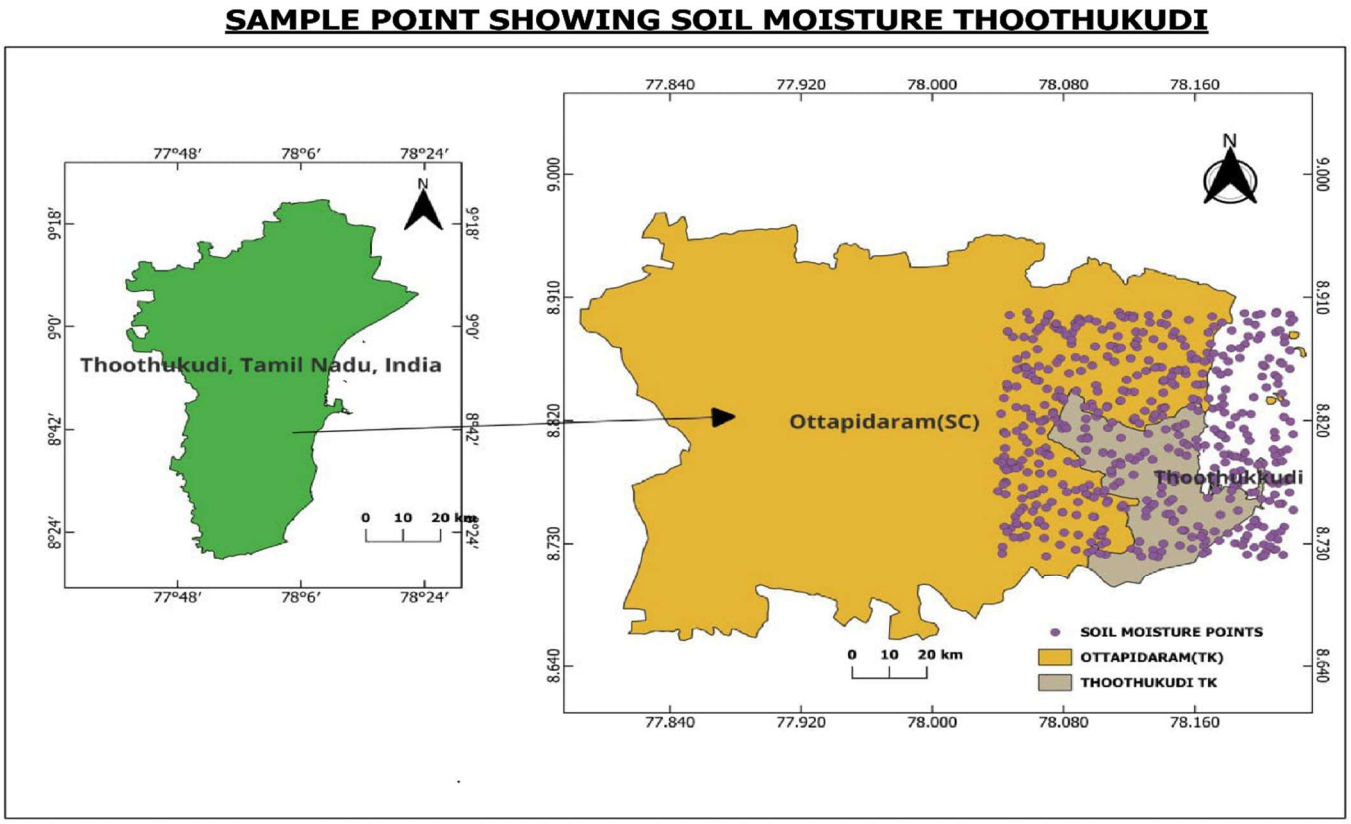


Fig 5.1 Sample Point Showing Soil Moisture Thoothukudi

The project combines different datasets like Sentinel-1 SAR imagery, soil class and soil moisture from Radar Vegetation Index (RVI) or field datasets. The green raster is the corrected backscatter layer from Sentinel-1 which helps to analyse the surface moisture whereas the gray points are the sample points or the extracted values of soil moisture. The soil raster is classified into different types of soils which helps in soil moisture retention. The CSV file that contains RVI and the geolocation is added as a layer which can be used to correlate the spatial relationship between the raster image datasets. The tools available in QGIS Processing Toolbox can perform advanced operations, for example, interpolation, raster statistics, and raster to vector conversion. The soil moisture variation will be helpful for research or planning departments to identify the spatial variation of the soil moisture and it is greatly helpful for agriculture or to monitor the drought condition.

5.3 GOOGLE EARTH ENGINE

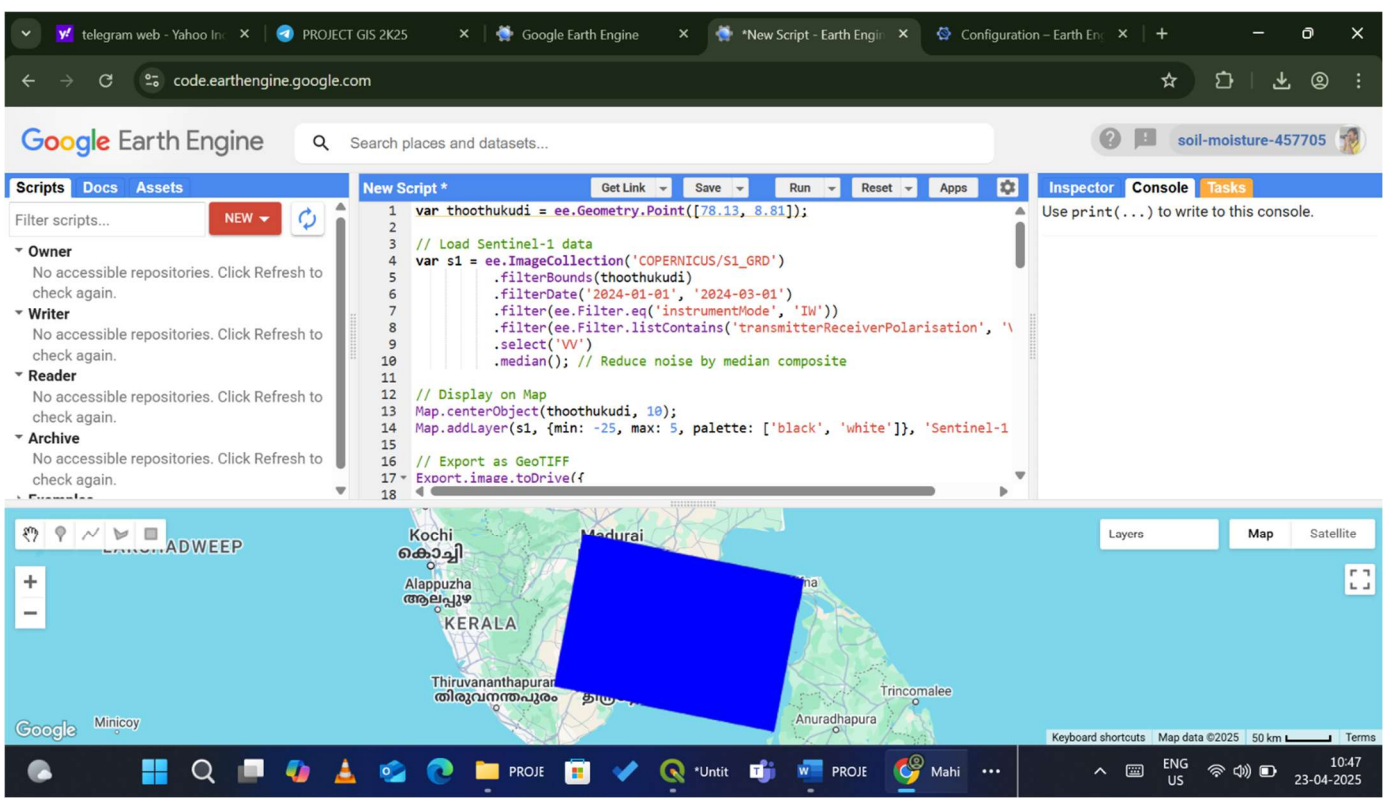


Fig 5.2 Google Earth Engine

A project workflow within Google Earth Engine (GEE) focused on extracting soil moisture-related data using Sentinel-1 satellite imagery for the Thoothukudi region.

The code employs JavaScript to designate a point of interest (thoothukudi = ee.Geometry.Point([78.13, 8.81])). It leverages the COPERNICUS/S1_GRD image collection, refines the dataset by area, timeframe (Oct to Dec 2024), mode (IW), and polarization (VV), and generates a median composite to address radar data noise. The product is a refined Sentinel-1 backscatter representation with information reflecting soil moisture levels. The refined information is showcased in a map using a black-and-white scheme, with the focus centered on the Thoothukudi location. Finally, the code packages the output for retrieval in the form of Geo TIFF files through Export.image.to drive, enabling downloading and inclusion in GIS software like QGIS. The sequence is a crucial phase in the soil moisture prediction process accomplished through Sentinel-1 satellite data analysis intended for enhanced spatial utilization.

5.4 VISUALISATION OF SOIL MOISTURE

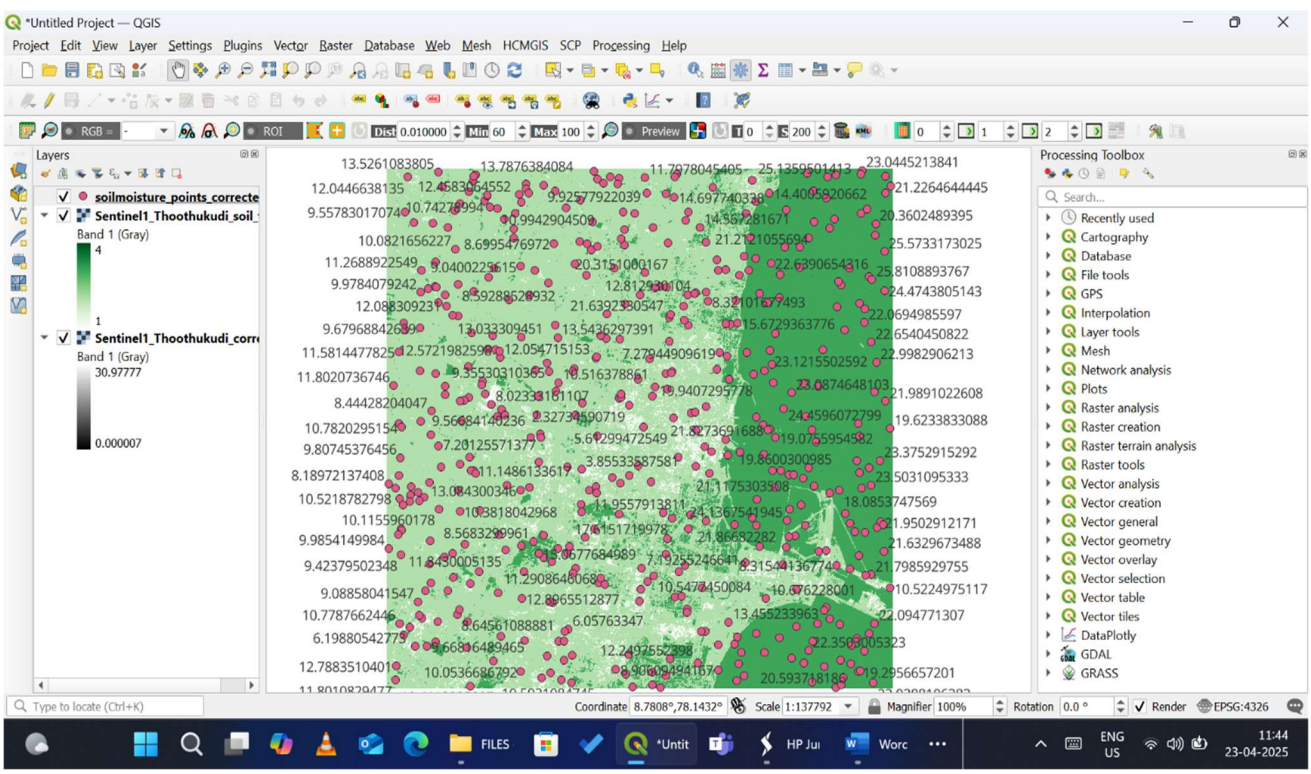


Fig 5.3 Visualisation of Soil Moisture

A progressed GIS visualization in QGIS, specializing in soil moisture evaluation inside the Thoothukudi area.More than one raster and vector layers are loaded, such as Sentinel1_Thoothukudi_corrected and Sentinel1_Thoothukudi_soil_type (grayscale Rasters), in addition to a vector point layer soil moisture_points_corrected. The map is densely populated with soil moisture point information that appears as magenta dots, overlaid with numerical labels to be had representation RVI (Radar Vegetation Index) or generated moisture values at the ones points. These are superimposed over the Radar derived image withinside the Sentinel-1, in which varying sunglasses of green and gray coloration depict the surface and vegetation in any other case. This advanced visualization allows spatial analysis of the moisture distribution, i.e., overlaying the radar reflectance records with the soil moisture point area data or the interpolated point records. The processing toolbox is open, which enables the person to get entry to the numerous gear and capabilities inclusive of the raster calculation, interpolation, orthe classification, for the analysis of the backscatter and moisture content material of the soil. This step is the fusion of the remote sensing and field data to be able to be used in the environmental or agriculture area.

5.5 DIGITAL ELEVATION MODEL THOOTHUKUDI:

The figure is a Digital Elevation Model (DEM) of Thoothukudi district, Tamil Nadu state, India. The DEM is a visual presentation of the terrain elevation throughout the district on a color gradient. Elevation ranges from -12 meters to 290 meters above mean sea level, with lower elevations presented as shades of red and orange, while higher elevations are indicated by yellow and white shades. The map has geographic coordinates along the boundary, a distance scale bar (0–16 km), and a north direction arrow. The district boundary has been prominently marked in black .This DEM is derived from Band 1 (Gray) data and will be helpful for different purposes like urban planning, hydrological modeling,

environmental studies, and disaster management in Thoothukudi.

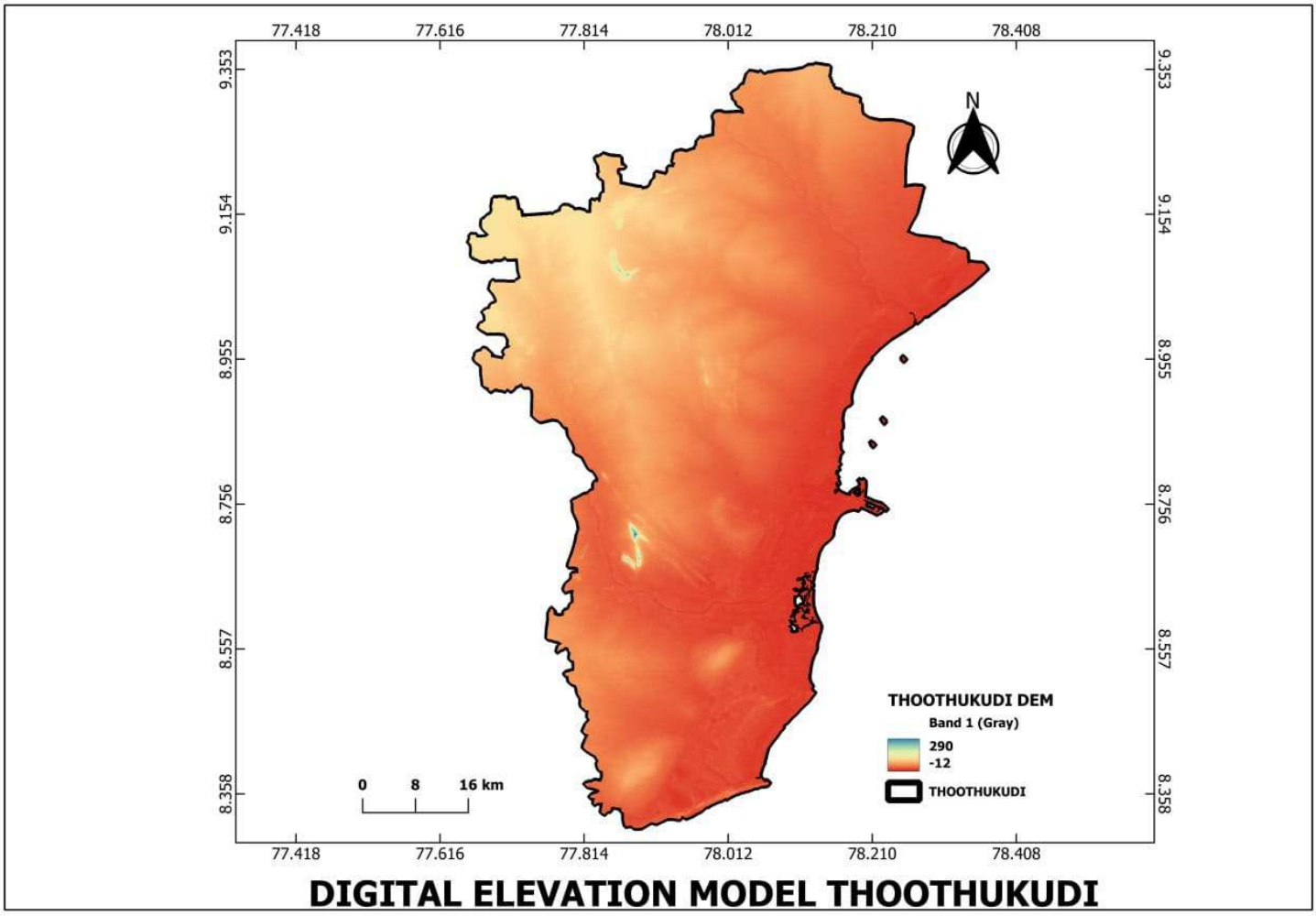


Fig 5.4 Digital Elevation Model Thoothukudi

5.6 SOIL MOISTURE INDEX THOOTHUKUDI :

The picture shows the Soil Moisture Index (SMI) map of Tamil Nadu's Thoothukudi overlaid on a southern India base map. The map indicates changes in the soil moisture of the area through a color gradation from green (high) to yellow (low). Bottom-left scale bar is used to indicate distances in miles (to a maximum of 50 miles), and bottom-right legend indicates soil moisture index values between -1 and 1.

Data is shown as a raster layer (float_raster), and SMI data color-coded assists in the identification of areas with various moisture contents. For instance, blue and green areas show regions of greater soil moisture content, whereas orange and yellow regions show drier soils. The overlay is inclined compared to the base map, evidently concentrating on Thoothukudi and the surrounding regions. The map can be utilized for planning agriculture, monitoring drought, and land management in the region. North direction is indicated with a compass rose at the top right for orientation.

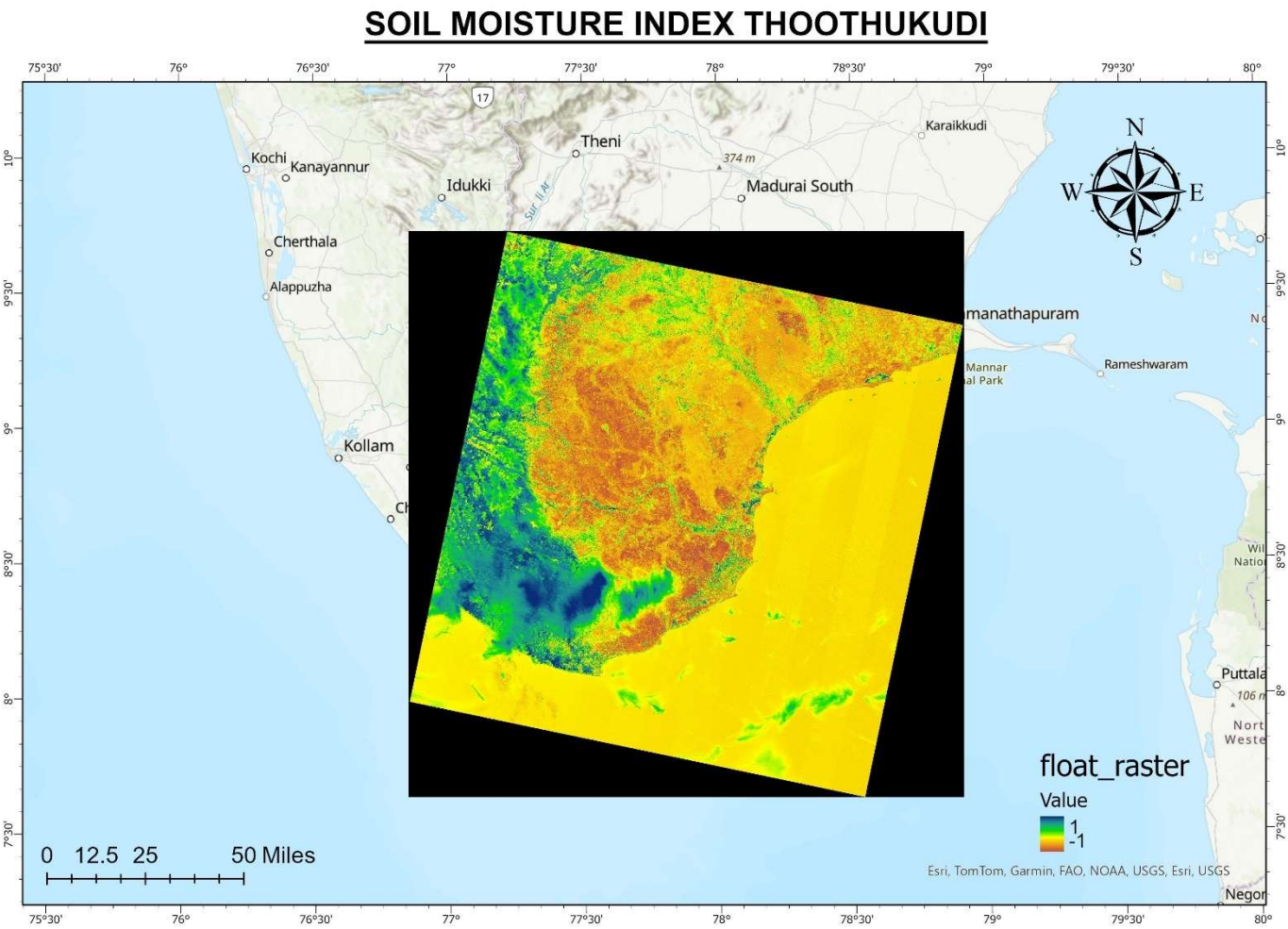


Fig 5.5 Soil Moisture Index Thoothukudi

5.7 TAMIL NADU SOIL MOISTURE MAP

The map is a Tamil Nadu Soil Moisture Map that shows soil moisture content over different districts of the state. The map employs a color-coded legend to depict five soil moisture classes:

1) Very dry (red) - Drought conditions, agriculture without irrigation.
2) Dry (orange)- Drought-resistant crops., Rainfall might be below average.
3) Medium (yellow)- crop types with minimal irrigation, moderate rainfall or irrigation support.
4) Wet (green) - Excellent for crop growth, Suggests good rainfall or presence of irrigation systems.
5) Very wet (blue) - flooding, waterlogging, or very heavy rainfall, crops due to risk of root rot and soil erosion.

Most of the districts, such as Thoothukudi (Tuticorin), Tirunelveli, Madurai, Trichy, and others, are categorized under the "Very dry" category, showing a large shortage of soil moisture. Few scattered pockets such as Nilgiris, Coimbatore, and Dharmapuri have medium to dry moisture. District boundaries are also indicated on the map along with a north arrow and scale bar (0–200 km) for spatial orientation. This map is important for evaluating drought conditions, irrigation planning, and agricultural decision-making based on the spatial pattern of soil moisture in Tamil Nadu.

Only a few isolated areas like Nilgiris, Coimbatore, and Dharmapuri show medium to dry moisture levels. The map also marks district boundaries and includes a north arrow and scale bar (0–200 km) for spatial reference. This map is crucial for assessing drought conditions, planning irrigation strategies, and making agricultural decisions based on the spatial distribution of soil moisture in Tamil Nadu.

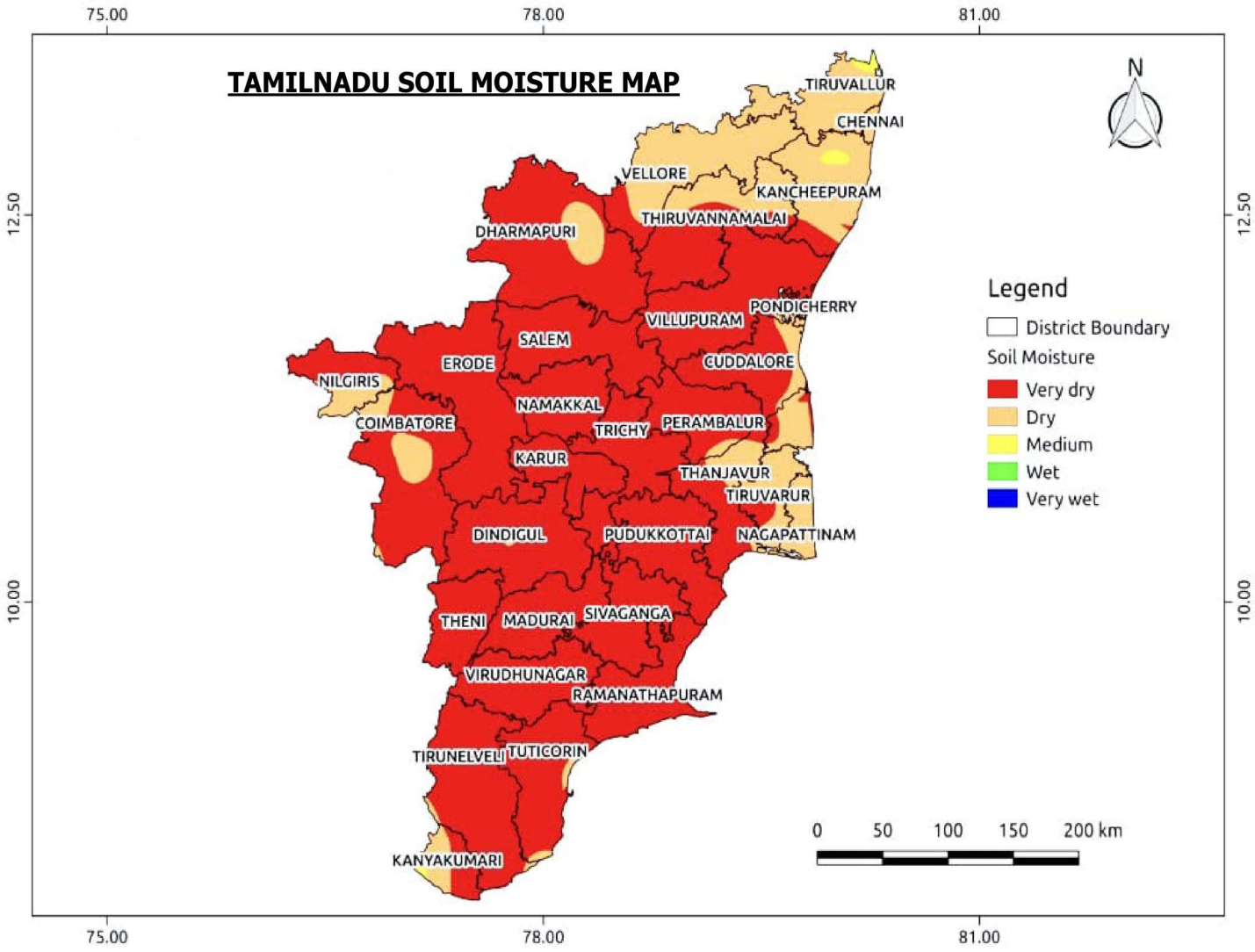


Fig 5.6 Tamil Nadu Soil Moisture Mapping

5.8 THOOTHUKUDI SOIL MOISTURE MAP :

The picture is captioned "Thoothukudi Soil Moisture Map", and it shows the spread of various soil textures in the Thoothukudi district of Tamil Nadu. The map incorporates a clear system of color-coding different soils, which are all key determiners of moisture content retention and suitability for agriculture. Going by the legend given

1) White marks - High moisture retention, but poor drainage,Waterlogged if over-irrigated.
2) Loamy Sand - Low water retention, drains quickly, Poor for crops that need consistent moisture.
3) Sandy green - Very low moisture holding capacity, Prone to drying out quickly.
4) Sandy yellow - Good moisture retention, but less drainage. Can crack when dry.
5) Sandy red - Moderate water retention, Good drainage, better than clay soils.
6) Sandy blue-green - holds moisture but drains well.

The map indicates that loamy sand and sandy clay loam prevail over most of the northern and central Thoothukudi area, with sand and sandy clay more heavily distributed in the southern and eastern areas. The occurrence of diverse soil textures within the district points to contrasting potentials for moisture storage and crop growth. The map contains a scale bar (kilometers), north arrow, and well-defined boundaries and hence is a useful tool for soil management, irrigation planning, and agricultural zoning for the area.

The map reveals that loamy sand and sandy clay loam dominate much of the northern and central Thoothukudi region, while sand and sandy clay are more concentrated in the southern and eastern parts. The presence of varied soil textures across the district indicates differing capabilities for moisture retention and crop cultivation. The map includes a scale bar (in kilometers), north arrow, and clearly marked boundaries, making it a valuable tool for soil management, irrigation planning, and agricultural zoning in the region.

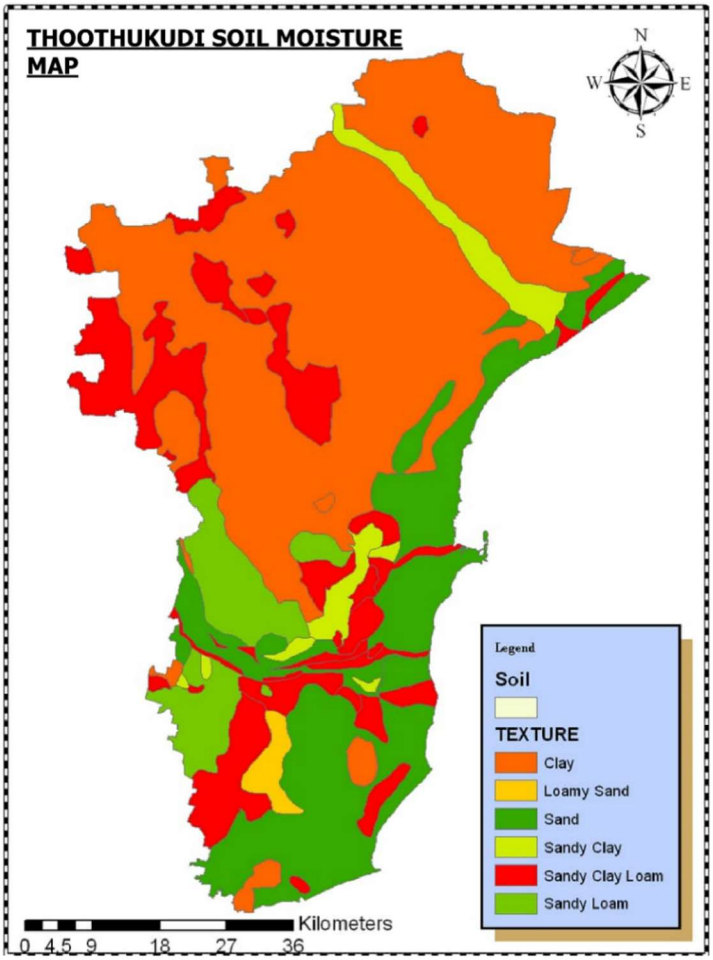


Fig 5.7 Thoothukudi Soil Moisture Map

5.9 LANDUSE LANDCOVER THOOTHUKUDI

The map entitled "Landuse Landcover (2024) Thoothukudi" presents the spatial distribution of various land use and land cover types of the Thoothukudi district, Tamil Nadu, India. The map classifies the surface features of the district with a colorful color-coded legend:

1) Built-up Area (red) - reflects human habitation and infrastructure.

2) Rangeland (orange) - reflects grazing or barren land.

3) Crops (yellow) - indicate agricultural lands under cultivation.

4) Trees (in green) - show forested or thickly vegetated areas.

5) Water (in blue) - shows water bodies like ponds, lakes, and rivers.

The map indicates that most of Thoothukudi is covered by cropland (yellow), especially in the central and northern parts. Wooded areas are grouped in the eastern and southern belts, while water bodies are scattered mainly in the southern and western parts. Urban areas are seen as small red clumps, mainly near urban areas and coastal districts.

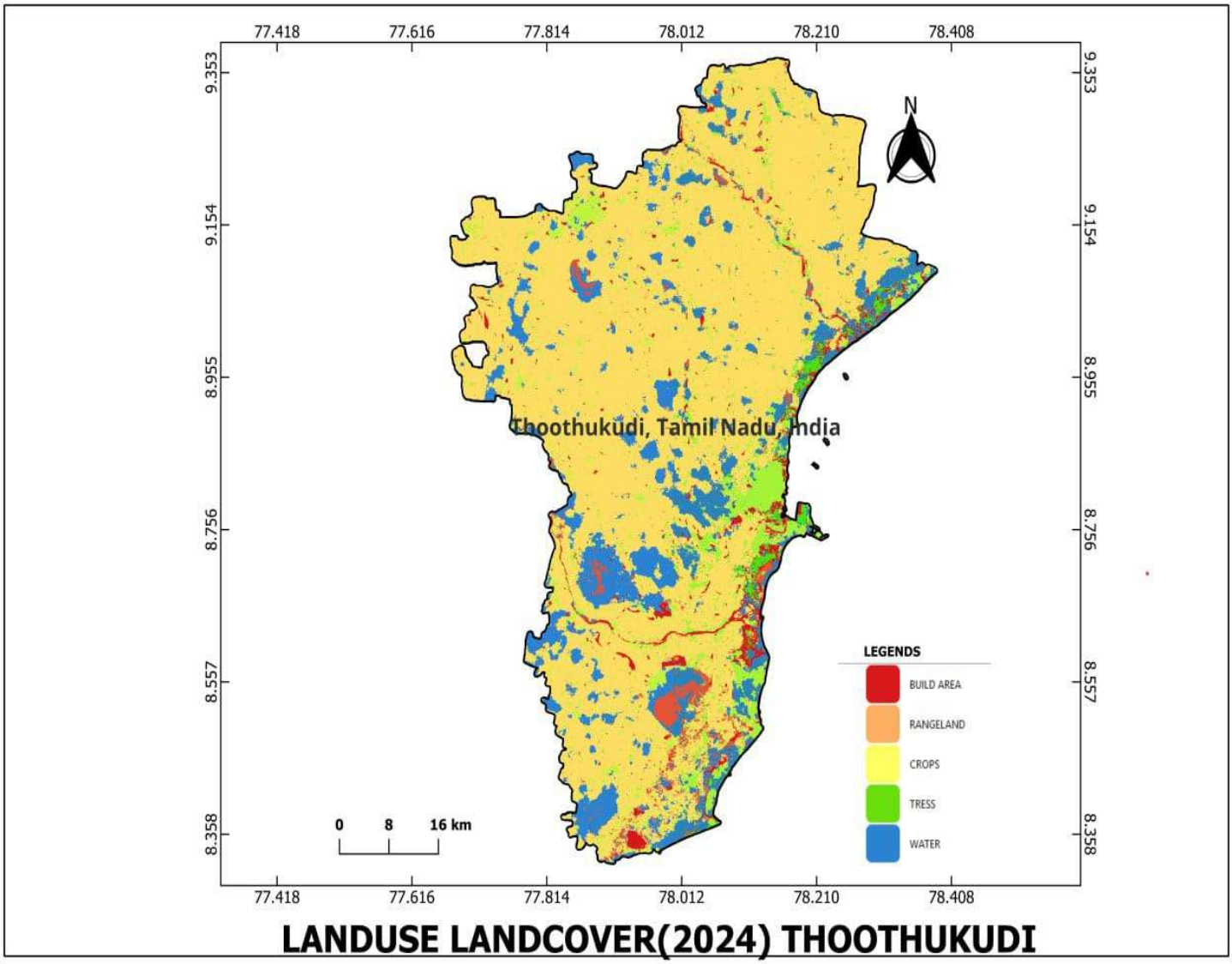


Fig 5.8 Landuse Landcover(2024) Thoothukudi

5.10 SOIL MOISTURE DENSITY THOOTHUKUDI

The map "Soil Moisture Density - Thoothukudi" displays the spatial distribution and heterogeneity of soil moisture in Thoothukudi district based on satellite remote sensing and field measurements. The map overlays two data bands: NDVI (Normalized Difference Vegetation Index) displayed in gray scales (vegetation density), and Moisture content, displayed in color gradients (green indicating high moisture, brown/orange indicating low).

Main features of the map are:

1) Black box – It indicated study area Thoothukudi district.

2) Sandy Red spots – ground-truthing locations where soil moisture values were physically measured or recorded.

3) Green – High soil moisture, Healthier, greener vegetation.

4) Red– Low moisture areas, Sparse or unhealthy vegetation.

5) The base map offers an explicit spatial context with latitude and longitude lines for accurate geographic referencing.

The map offers informative insights into the health of the soil, aiding in agricultural planning, irrigation management, and drought monitoring. It also points out areas that need water conservation work and the zones with potentially improved crop yield because of sufficient moisture content.

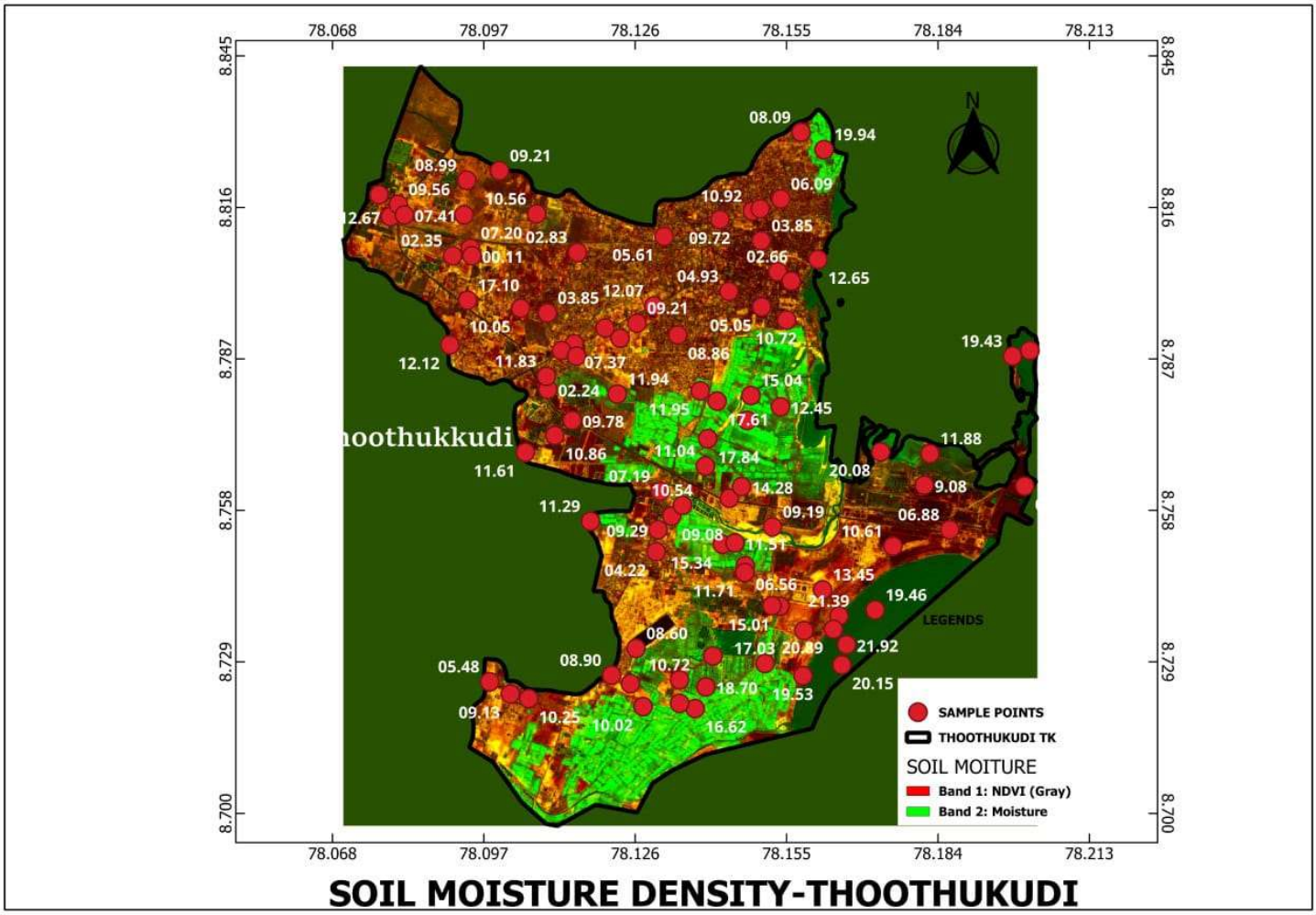


Fig 5.9 Soil Moisture Density Thoothukudi

CHAPTER 6
CONCLUSION

Soil moisture content is of great importance in assessing soil plant water interaction and studying the climate change. Soil Moisture is also the key finder for the ground water level. The findings underscore the importance of adopting advanced geospatial techniques for environmental monitoring and resource management. Implementing such GIS-based soil moisture mapping can significantly contribute to sustainable agricultural practices, efficient water management, and disaster preparedness in Thoothukudi district.

Soil moisture map for Thoothukudi district using GIS technology, with a special focus on the Thoothukudi and Ottapidaram regions. Field data was collected from 100 sampling points, each covering a 300-meter radius. Using spatial interpolation methods in a GIS environment, a continuous map of soil moisture distribution was produced, helping to visualize how moisture levels vary across the district. Three raster cells showed negative values, which often occur in areas with very low reflectance. This can happen in locations like water bodies, saturated soils, dense vegetation, or shaded areas.

From the above study it is concluded that remotely sensed data can be used for the mapping of soil moisture as it is less time and resource consuming. By employing satellite-derived datasets and GIS techniques, The study has provided spatial insights into soil moisture distribution, which are important for informed decision-making in agriculture and water resource management. This paper outlines the basic principles of satellite-based techniques for soil moisture mapping.

REFERENCES

1. Alihsan Sekertekin, Ayan Murat Marangoz, Saygin Abdikan, "SOIL MOISTURE MAPPING USING SENTINEL-1A SYNTHETIC APERTURE RADAR DATA"Vol:5 Issue:2 July 2018.
2. NazmusSazib, Iliana Mladenova and John Bolten "LEVERAGING THE GOOGLE EARTH ENGINE FOR DROUGHT ASSESSMENT USING GLOBAL SOIL MOISTURE DATA" 11 August 2018
3. Sachin Sutariya, Ankur Hirapara, Momin Meherbanali, M.K. Tiwari, Vijay Singh, Manik H. Kalubarme." SOIL MOISTURE ESTIMATION USING SENTINEL-1 SAR DATA AND LAND SURFACE TEMPERATURE IN PANCHMAHAL DISTRICT, GUJARAT STATE", Vol: 8 Issue:1 25 Nov 2020.
4. Felix Greifeneder , Claudia Notarnicola and Wolfgang Wagner," A MACHINE LEARNING-BASED APPROACH FOR SURFACE SOIL MOISTURE ESTIMATIONS WITH GOOGLE EARTH ENGINE " 27 May 2021.
5. Jinyang Du , John S. Kimball , Rajat Bindlish , Jeffrey P. Walker , Jennifer D. Watts," LOCAL SCALE (3-M) SOIL MOISTURE MAPPING USING SMAP AND 2 PLANET SUPER DOVE ", June 10,2022.
6. Janani N, Balaji Kannan, Nagarajan K, Thiyagarajan G, Duraisamy M R," TREND ANALYSIS AND VARIABILITY OF SATELLITE-BASED SOIL MOISTURE DATA FOR THE LOWER BHAVANI BASIN, TAMIL NADU USING GOOGLE EARTH ENGINE" May 11,2023.
7. Abolghasem Akbari, MajidRajabi Jagharg Azizan AbuSamah Jonathan PeterCox|, Mojtaba Gholamzadeh , Alireza Araghi, PatriciaM.Saco, "UTILIZATION OF THE GOOGLE EARTH ENGINE FOR THE EVALUATION OF DAILY SOI TEMPERATURE DERVED FROM GLOBAL LAND DATA ASSIMILATION SYSTEM IN TWO DIFFERENT DEPTHS OVERASE MIARID REGION" 27 June 2024.

8. Shadia A. Alzurqani Shawn Jackson , Hamdi A. Zurqani , Don White Jr., Kathleen Bridges," GOOGLE EARTH ENGINE APPLICATION FOR MAPPING AND MONITORING DROUGHT PATTERNS AND TRENDS: A CASE STUDY IN ARKANSAS, USA", 19 Oct 2024.
9. Zhenghao Li, Qiangqiang Yuan & Xin Su," HIGH-SPATIAL-RESOLUTION SURFACE SOIL MOISTURE RETRIEVAL USING THE DEEP FOREST MODEL IN THE CLOUD ENVIRONMENT OVER THE TIBETAN PLATEAU",15 Feb 2024.
10. A.Gorelick, M.Hancher, M.Dixon, "REMOTE SENSING OF SOIL MOISTURE USING GOOGLE EARTH ENGINE", Remote Sensing of Environment, Volume:202,2017, Edition :2017.
11. M.Bauer-Marschallinger, W.Wagner, T.Reimer, "GLOBAL SOIL MOISTURE ESTIMATION USING GOOGLE EARTH ENGINE AND SENTINEL-1 DATA", Remote Sensing, Volume 11(19), 2019, Volume 11, Issue 19.
12. T.Rezaei, M.Shamsipour, "SOIL MOISTURE RETRIEVAL USING SENTINEL-1 SAR AND GOOGLE EARTH ENGINE", Journal of Hydrology, Volume: 594, 2021, Edition: Volume 594.
13. Z.Li,H. Zhang," MONITORING SOIL MOISTURE USING GOOGLE EARTH ENGINE: A CASE STUDY OF THE LOESS PLATEAU", Sustainability, Volume: 12(13), 2020, Edition: Volume 12, Issue 13.
14. S.Djamai, S.Magagi, R.Roy," SPATIO-TEMPORAL ANALYSIS OF SOIL MOISTURE USING GOOGLE EARTH ENGINE IN AN AGRICULTURAL REGION", Computers and Electronics in Agriculture, Volume: 177, 2020, Edition: Volume 177.

15. S.Joshi, M.R.Khan, A.Kumar, "A MACHINE LEARNING APPROACH TO SOIL MOISTURE ESTIMATION USING GEE AND SENTINEL-1", Geocarto International, Volume: 38(2), 2023, Edition: Volume 38, Issue 2.
16. E.Beck, K.Heuvelink, J.van den Bosch, "SOIL MOISTURE DYNAMICS DERIVED FROM RADAR BACKSCATTER USING GOOGLE EARTH ENGINE", International Journal of Remote Sensing, Volume: 42(1), 2021, Edition: Volume 42, Issue 1.
17. M.Sharma, R.Rawat, D.Singh, "EVALUATING SOIL MOISTURE RETRIEVAL FROM SENTINEL-1 USING GOOGLE EARTH ENGINE", Journal: Sensors, Volume: 20(18), 2020, Edition: Volume 20, Issue 18.
18. K.Wang, F.Zhang, Y.Liu, "INTEGRATION OF MODIS AND SMAP DATA FOR SOIL MOISTURE ESTIMATION USING GEE", Remote Sensing, Volume: 10(8), 2018, Edition: Volume 10, Issue 8.
19. J.Brown, M.G.Anderson, "SOIL MOISTURE MAPPING USING OPTICAL AND RADAR SATELLITE DATA WITH GOOGLE EARTH ENGINE", Journal of Remote Sensing & GIS, Volume: 6(4), 2017, Edition: Volume 6, Issue 4.
20. L.Tian, Y.Wang, M.Zhang, "USING GEE FOR HIGH-RESOLUTION SOIL MOISTURE ESTIMATION IN MOUNTAINOUS TERRAIN", Remote Sensing Applications: Society and Environment, Volume: 25, 2022, Edition: Volume 25.
21. A.George, K.Mathew, S.Iyer, "SOIL MOISTURE VARIATION ANALYSIS USING SENTINEL-1 AND GEE IN TROPICAL CLIMATES", Arabian Journal of Geosciences, Volume: 14(11), 2021, Edition: Volume 14, Issue 11.
22. N.Prakash, K.R.Singh, "REMOTE SENSING-BASED SOIL MOISTURE ESTIMATION USING TIME-SERIES SAR DATA ON GEE", Advances in Space Research, Volume: 68(2), 2021, Edition: Volume 68, Issue 2.

APPENDIX

Code For Arc Pyimport arcpy

```
from arcpy import mapping

# Set the MXD file (current ArcMap document)
mxd = mapping.MapDocument("CURRENT")
df = mapping.ListDataFrames(mxd)[0]  # Use the first DataFrame

# Define file paths (modify these paths if needed)
soil_moisture_tiff = r"C:\path\to\Sentinel1_Thoothukudi_corrected.tif"
soil_type_tiff = r"C:\path\to\Sentinel1_Thoothukudi_soil_type.tif"
output_pdf = r"C:\path\to\Thoothukudi_Soil_Moisture_Map.pdf"

# Add Soil Moisture Raster Layer
moisture_layer = mapping.Layer(soil_moisture_tiff)
mapping.AddLayer(df, moisture_layer, "BOTTOM")

# Add Soil Type Raster Layer
type_layer = mapping.Layer(soil_type_tiff)
mapping.AddLayer(df, type_layer, "TOP")

# Adjust legend properties
legend = mapping.ListLayoutElements(mxd, "LEGEND_ELEMENT")[0]
legend.title = "Soil Moisture & Types"
legend.autoAdd = True
```

```
# Insert North Arrow
north_arrow = mapping.ListLayoutElements(mxd, "MAPSURROUND_ELEMENT")[0]
north_arrow.elementPositionX = 1.0
north_arrow.elementPositionY = 7.5

# Insert Scale Bar
scale_bar = mapping.ListLayoutElements(mxd, "MAPSURROUND_ELEMENT")[1]
scale_bar.elementPositionX = 5.0
scale_bar.elementPositionY = 1.0

# Insert Title
title = mapping.ListLayoutElements(mxd, "TEXT_ELEMENT")[0]
title.text = "Thoothukudi Soil Moisture"
title.elementPositionX = 3.0
title.elementPositionY = 8.5
title.fontSize = 18
title.bold = True

# Enable Grid (Graticule)
grid = mapping.ListDataFrames(mxd)[0].grid
if not grid:
    arcpy.AddMessage("Adding Grid...")
    grid = mapping.Grid()
    grid.name = "Lat-Long Grid"
    grid.setCoordinateSystem("GCS_WGS_1984")
    grid.enabled = True
df.spatialReference = arcpy.SpatialReference(4326)  # WGS 1984
```

```
# Refresh and Export the Map
mapping.ExportToPDF(mxd, output_pdf)
arcpy.AddMessage("Map exported successfully: " + output_pdf)

# Clean up
del mxd
code for GEE Final Project   // Define Thoothukudi region
var thoothukudi = ee.Geometry.Point([78.13, 8.81]);

// Load Sentinel-1 data
var s1 = ee.ImageCollection('COPERNICUS/S1_GRD')
        .filterBounds(thoothukudi)
        .filterDate('2024-01-01', '2024-03-01')
        .filter(ee.Filter.eq('instrumentMode', 'IW'))
        .filter(ee.Filter.listContains('transmitterReceiverPolarisation', 'VV'))
        .select('VV')
        .median(); // Reduce noise by median composite

// Display on Map
Map.centerObject(thoothukudi, 10);
Map.addLayer(s1, {min: -25, max: 5, palette: ['black', 'white']}, 'Sentinel-1 VV');

// Export as GeoTIFF
Export.image.toDrive({
  image: s1,
  description: 'Sentinel1_Thoothukudi',
  scale: 10,
```

```
region: thoothukudi.buffer(10000).bounds(),
  fileFormat: 'GeoTIFF'});

// Define Thoothukudi region
var thoothukudi = ee.Geometry.Point([78.13, 8.81]);

// Load Sentinel-1 Data (VV & VH Polarization)
var s1 = ee.ImageCollection('COPERNICUS/S1_GRD')
        .filterBounds(thoothukudi)
        .filterDate('2024-01-01', '2024-03-01')
        .filter(ee.Filter.eq('instrumentMode', 'IW'))
        .filter(ee.Filter.listContains('transmitterReceiverPolarisation', 'VV'))
        .filter(ee.Filter.listContains('transmitterReceiverPolarisation', 'VH'))
        .select(['VV', 'VH'])
        .median(); // Reduce noise

// Compute Radar Vegetation Index (RVI)
var rvi = s1.expression('(4 * VH) / (VV + VH)', {
  'VV': s1.select('VV'),
  'VH': s1.select('VH')
}).rename('RVI');

// Visualization parameters (High RVI = Wet soil, Low RVI = Dry soil)
var visParams = {min: 0, max: 1, palette: ['brown', 'yellow', 'green', 'blue']};

// Display on the map
Map.centerObject(thoothukudi, 10);
```

```
Map.addLayer(rvi, visParams, 'Soil Moisture (RVI)');

// Export RVI Map
Export.image.toDrive({
  image: rvi,
  description: 'Soil_Moisture_Thoothukudi',
  scale: 10,
  region: thoothukudi.buffer(10000).bounds(),
  fileFormat: 'GeoTIFF'
});
// Define Thoothukudi region
var thoothukudi = ee.Geometry.Point([78.13, 8.81]);

// Load Sentinel-1 Data (VV & VH Polarization)
var s1 = ee.ImageCollection('COPERNICUS/S1_GRD')
        .filterBounds(thoothukudi)
        .filterDate('2024-01-01', '2024-03-01')
        .filter(ee.Filter.eq('instrumentMode', 'IW'))
        .filter(ee.Filter.listContains('transmitterReceiverPolarisation', 'VV'))
        .filter(ee.Filter.listContains('transmitterReceiverPolarisation', 'VH'))
        .select(['VV', 'VH'])
        .median();

// Compute Radar Vegetation Index (RVI)
var rvi = s1.expression('(4 * VH) / (VV + VH)', {
  'VV': s1.select('VV'),
  'VH': s1.select('VH')
```

```
}).rename('RVI');

// Define Export Region (10 km buffer around Thoothukudi)
var exportRegion = thoothukudi.buffer(10000).bounds();

// Export Image as GeoTIFF
Export.image.toDrive({
  image: rvi,
  description: 'Soil_Moisture_Thoothukudi',
  scale: 10,
  region: exportRegion,
  fileFormat: 'GeoTIFF'
});
```